Pascal's hexagon theorem over an arbitrary field, formalised in Lean 4

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Abstract

We formalise Pascal's hexagon theorem in Lean 4 with Mathlib. The statement covers an arbitrary conic over an arbitrary field: six points on the zero set of a nonzero ternary quadratic form. It places no restriction on the characteristic and allows degenerate conics. The only hypotheses are the ones needed for the three intersection points of opposite sides to be defined. Pappus's theorem follows as the case of a line pair, and Brianchon's dual-conic formulation as the projective dual. The formal statement uses Mathlib's projective plane $\mathbb{P}(K^3)$, its cross-product construction of lines and intersection points, and its notion of collinearity. We also prove the lemmas a reader needs to check that this statement means what it says. The proof rests on a single polynomial identity, valid over any commutative ring: the triple product of the three Pascal points is, up to sign, the 6×6 determinant of the degree-two Veronese images of the six vertices. The mathematics is classical. As far as we know, this is the first Lean formalisation, without `sorry` or added axioms, for an arbitrary nonzero ternary quadratic form over an arbitrary field. It extends the real projective statements formalised in HOL Light and Mizar to arbitrary fields. Those statements assume 17 or 18 non-collinearity conditions on the vertices; we replace them by distinctness of consecutive vertices and of opposite sides. The development has 32 named theorems in about 670 lines. A checking script, tested against deliberately tampered copies of the source, confirms that each of them uses only Lean's standard axioms. Version 2 adds Mathlib-only restatements with explicit non-degenerate witnesses, and a comparison with independently written formalisations of the textbook statements. That comparison proves, outside the main development, Brianchon's theorem for tangent lines in characteristic $\neq 2$, the five-point (tangent) form of Pascal's theorem and its converse.

1 Introduction

Pascal's theorem (1640) is the statement that the three points where opposite sides of a hexagon inscribed in a conic meet are collinear. It is number 28 on Wiedijk's list of 100 theorems [10]. The list records formalisations in HOL Light, Mizar and Rocq. Mathlib's copy of the list has no Lean declaration for it [7].

We prove the following.

Theorem 1 (Pascal)**.** *Let K be a field and Q a nonzero quadratic form on K^3. Let A, B, C, D, E, F be points of the projective plane $\mathbb{P}^2(K)$ with $Q(A) = \cdots = Q(F) = 0$. Suppose consecutive vertices are distinct, $A \neq B$, $B \neq C$, …, $F \neq A$, and opposite sides are distinct lines, $AB \neq DE$, $BC \neq EF$, $CD \neq FA$. Then the points $AB \cap DE$, $BC \cap EF$ and $CD \cap FA$ are collinear.*

No assumption is made on the characteristic of K, on the nondegeneracy of Q, or on nonconsecutive vertices. The hypotheses are exactly what is needed for the six sides to be lines and for the three intersection points to exist. If $Q = \ell_1\ell_2$ is a product of two linear forms, Theorem 1 is Pappus's theorem (Corollary 12).

Contributions.

- A Lean 4 formalisation of Theorem 1 (`PascalHexagon.pascal`). It is stated with Mathlib's `Projectivization`, `Projectivization.cross` and `Projectivization.IsCollinear`.
- A vector form over any integral domain with no distinctness hypotheses at all (Theorem 5), and the underlying identity over any commutative ring (Proposition 2).
- Lemmas that justify the formal statement (Section 5). They show that `cross A B` is the unique line through A and B, that `cross L M` is the unique point on L and M, that collinearity is equivalent to the vanishing of a determinant, and that lying on the conic does not depend on the chosen representative. We also give a version whose conclusion does not mention `cross` at all.
- Pappus's theorem as a corollary, and Brianchon's dual-conic formulation as the projective dual (Section 6).
- A tamper-tested checking script that confirms the axiom footprint of every named theorem (Section 8).
- In version 2 (Section 8.1): Mathlib-only restatements of the main results, with explicit nondegenerate witnesses for statements with hypotheses, and a comparison with a blind formalisation of the textbook statements. The comparison proves all 20 textbook statements, among them Brianchon's theorem for tangent lines (characteristic $\neq 2$), the tangent form of Pascal's theorem and the Braikenridge–Maclaurin converse.

Availability. The code is at `https://github.com/jpbald93/pascal-hexagon`. Line references in this paper are to commit `726da4d`. The toolchain is Lean `v4.33.1`, and Mathlib is pinned in `lake-manifest.json`. The code is released under Apache 2.0. Version 2 of this paper adds the files `PascalHexagon/L14.lean` and `PascalHexagon/L15Classical.lean` (Section 8.1) in a new commit, `be35ffaa75a324131a710d3287b3d703408ad4a0`. In that commit the root import file and `gate.sh` gain entries for them, and `gate.sh` also gets a portability fix (Section 8.1). The earlier statements and files are unchanged, so the line references hold there as well. Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `Pascal-Brianchon` of the repository `https://github.com/jpbald93/LEAN-verification`.

2 Conventions and the Mathlib interface

We write vectors of K^3 as `Fin 3 → K`. The bracket of three vectors is $[u, v, w] = u \cdot (v \times w) = \det(u, v, w)$. Mathlib provides the cross product `u ×₃ v` and the identity $[u, v, w] = \det(u, v, w)$ (`triple_product_eq_det`).

The projective plane is Mathlib's `ℙ K (Fin 3 → K)`. Its points are nonzero vectors up to nonzero scalars. `P.rep` is a chosen representative, and `mk K v hv` is the point of a nonzero vector v. Mathlib

Lean name (namespace `PascalHexagon`)	In this paper	Source
`pascalDet_eq_pascalBracket`	Proposition 2(a)	`Bracket.lean`
`det_verMatrix_eq_neg_pascalBracket`	Proposition 2(b)	`Veronese.lean`
`pascalDet_eq_zero_of_quadEval`	Lemma 3	`Conic.lean`
`apply_eq_quadEval`	Lemma 4	`Conic.lean`
`coeffs_ne_zero`	Lemma 4	`Conic.lean`
`pascal_quadraticMap`	Theorem 5	`Conic.lean`
`pascal_matrix`	Remark 6	`Conic.lean`
`pascalDet_standardConic`	Remark 7	`StandardConic.lean`
`isCollinear_mk_iff`	Lemma 8	`Correctness.lean`
`mem_line_cross`	Lemma 9(a)	`Correctness.lean`
`cross_mem_lines`	Lemma 9(a)	`Correctness.lean`
`eq_cross_of_orthogonal`	Lemma 9(b)	`Correctness.lean`
`eq_cross_of_orthogonal'`	Lemma 9(b), dual	`Correctness.lean`
`cross_cross_spec`	Lemma 9(c)	`Correctness.lean`
`onConic_mk_iff`	Lemma 11	`Correctness.lean`
`pascal`	Theorem 1	`Projective.lean`
`pascal'`	Theorem 1, with `OnConic`	`Correctness.lean`
`pascal_of_isCollinear`	Theorem 10	`Correctness.lean`
`linePair_ne_zero`	Corollary 12, proof	`Correctness.lean`
`pappus`	Corollary 12	`Correctness.lean`
`brianchon`	Theorem 13	`Brianchon.lean`
`brianchon_concurrent`	Theorem 13, concurrency	`Brianchon.lean`

Table 1: The main formal statements. `gate.sh` checks these 22 theorems and 10 supporting lemmas, 32 theorems in all.

identifies lines with points of the same space via the dot product: `P.orthogonal L` means $P \cdot L = 0$, which we read as "P lies on L". For distinct points $A \neq B$, Mathlib's `cross A B` is the point represented by $A \times B$. It serves both as the line through two points and as the intersection point of two lines. Collinearity is Mathlib's `IsCollinear`: a set of points is collinear if it lies in a subspace of dimension at most 2. Mathlib's definition of `cross` requires `[DecidableEq K]`, so the statements that mention `cross` carry this hypothesis. It is harmless in classical logic.

A conic is the zero set of a nonzero `QuadraticForm K (Fin 3 → K)`. Here "nonzero" means nonzero as a quadratic form, which is the same as having a nonzero coefficient vector (Lemma 4). We write `OnConic Q P` for $Q(\texttt{P.rep}) = 0$.

3 The key identity

For vectors $A, \ldots, F \in K^3$, define the *Pascal determinant*

$$\pi(A, \ldots, F) = [(A \times B) \times (D \times E),\ (B \times C) \times (E \times F),\ (C \times D) \times (F \times A)].$$

If $A \neq B$ and $D \neq E$ as projective points and $AB \neq DE$, then $(A \times B) \times (D \times E)$ represents $AB \cap DE$, and similarly for the other two points. So Pascal's theorem says that π vanishes. Let $\nu(x, y, z) = (x^2, y^2, z^2, xy, yz, zx)$ be the degree-two Veronese map, and let $V(A, \ldots, F)$ be the 6×6 matrix with rows $\nu(A), \ldots, \nu(F)$.

Proposition 2. *Over any commutative ring, for all $A, \ldots, F \in K^3$:*

(a) $\pi(A,\dots,F) = \beta(A,\dots,F)$*, where* β *is the bracket polynomial*

$$\beta = [ABE][BCF][CDA][DEF] - [ABE][BCF][CDF][DEA] + [ABE][BCE][CDF][DFA] - [ABD][BCE][CDF][EFA];$$

(b) $\det V(A,\dots,F) = -\beta(A,\dots,F)$.

Hence $\pi = -\det V$.

Proof. (a) Apply the vector triple product identity $(u \times v) \times w = (u \cdot w)v - (v \cdot w)u$ to each of the three Pascal points. This gives, for instance,

$$(A \times B) \times (D \times E) = [A, D, E]B - [B, D, E]A.$$

Expanding the bracket of the three resulting vectors by multilinearity, and cancelling terms with a repeated argument, gives an expression in brackets. Its equality with β is a polynomial identity in the 18 coordinates. (b) This is a polynomial identity in the 18 coordinates. □

Formally, (a) is proved by rewriting with Mathlib's `cross_cross_eq_smul_sub_smul` and then calling `ring` on the coordinate expansion. (b) is proved by Laplace expansion along the first row, followed by `ring`. Neither proof is pretty. Each takes about 80–95 seconds and 2.5 GB of memory to check. They are, however, short, and they need no hypotheses. Brackets for the conic condition are classical (see e.g. [9]). HOL Light's formalisation also goes through a bracket condition (Section 7).

Lemma 3. *Let* K *be an integral domain and* $c \in K^6$ *nonzero. If* $\nu(v) \cdot c = 0$ *for* $v = A,\dots,F$*, then* $\pi(A,\dots,F) = 0$.

Proof. $V(A,\dots,F)\,c = 0$ with $c \neq 0$, so $\det V = 0$, and $\pi = -\det V = 0$ by Proposition 2. □

Lemma 4. *For every quadratic map* Q *on* K^3 *over a commutative ring,* $Q(v) = \nu(v) \cdot c_Q$*, where*

$$c_Q = \big(Q(e_1), Q(e_2), Q(e_3),\, b_Q(e_1,e_2),\, b_Q(e_2,e_3),\, b_Q(e_3,e_1)\big)$$

and $b_Q(x,y) = Q(x+y) - Q(x) - Q(y)$ *is the polar form. If* $Q \neq 0$*, then* $c_Q \neq 0$.

Theorem 5 (vector form)**.** *Let* K *be an integral domain and* $Q \neq 0$ *a quadratic map on* K^3*. If* $A,\dots,F \in K^3$ *satisfy* $Q(A) = \cdots = Q(F) = 0$*, then* $\pi(A,\dots,F) = 0$.

Proof. Lemmas 4 and 3. □

```
theorem pascal_quadraticMap [IsDomain K] {Q : QuadraticMap K (Fin 3 → K) K} (hQ : Q ≠ 0)
    {A B C D E F : Fin 3 → K}
    (hA : Q A = 0) (hB : Q B = 0) (hC : Q C = 0) (hD : Q D = 0) (hE : Q E = 0) (hF : Q F
    = 0) :
    ((A ×₃ B) ×₃ (D ×₃ E)) ·ᵥ ((B ×₃ C) ×₃ (E ×₃ F)) ×₃ ((C ×₃ D) ×₃ (F ×₃ A)) = 0
```

Theorem 5 has no distinctness hypotheses at all. When the configuration is degenerate, for example $A = B$, some Pascal point is the zero vector and the statement holds trivially. The content of Pascal's theorem lies in the nondegenerate case, which the projective form below isolates.

Remark 6. Many texts describe a conic by a symmetric matrix M with $v^{\mathsf{T}}Mv = 0$. The formal version `pascal_matrix` is stated over a field K with $2 \neq 0$: if $M \neq 0$ is symmetric and $v^{\mathsf{T}}Mv = 0$ for $v = A,\dots,F$, then $\pi(A,\dots,F) = 0$. The hypothesis $2 \neq 0$ is needed. In characteristic 2, $v^{\mathsf{T}}Mv$ sees only the diagonal of M, so a nonzero symmetric matrix can give the zero form. This is why we state the main results with quadratic forms.

Remark 7. The development also contains a direct proof for the standard parametrised conic $xz = y^2$, with points (s^2, st, t^2) (`pascalDet_standardConic`). It holds for all parameters s, t over any commutative ring, and it is not used in the main proof.

4 The projective statement

```
theorem pascal {Q : QuadraticForm K (Fin 3 → K)} (hQ : Q ≠ 0)
    {A B C D E F : ℙ K (Fin 3 → K)}
    (hQA : Q A.rep = 0) (hQB : Q B.rep = 0) (hQC : Q C.rep = 0) (hQD : Q D.rep = 0)
    (hQE : Q E.rep = 0) (hQF : Q F.rep = 0)
    (hAB : A ≠ B) (hBC : B ≠ C) (hCD : C ≠ D) (hDE : D ≠ E) (hEF : E ≠ F) (hFA : F ≠ A
    )
    (h₁ : cross A B ≠ cross D E) (h₂ : cross B C ≠ cross E F) (h₃ : cross C D ≠ cross F
    A) :
    IsCollinear ({cross (cross A B) (cross D E), cross (cross B C) (cross E F),
      cross (cross C D) (cross F A)} : Set (ℙ K (Fin 3 → K)))
```

The field K carries `[Field K] [DecidableEq K]`. The proof unfolds `cross` on representatives. Theorem 5 then shows that the triple product of the representatives of the three points vanishes, and Lemma 8(b) turns this into collinearity.

Lemma 8. *For nonzero $u, v, w \in K^3$:*

(a) if $\{[u], [v], [w]\}$ is collinear, then $[u, v, w] = 0$;

(b) if $[u, v, w] = 0$, then $\{[u], [v], [w]\}$ is collinear.

Part (b) is `isCollinear_mk_of_triple_product_eq_zero`. It is used in the proof of `pascal`. Part (a) is its converse, and the two together give `isCollinear_mk_iff`. Part (a) is not needed for Pascal's theorem. We include it so that "collinear" in the formal statement can be read as the familiar determinant condition.

5 Does the statement mean what it says?

A reader of `pascal` must trust two definitions: that `cross A B` is the line AB, and that `cross (cross A B) (cross D E)` is the point $AB \cap DE$. Mathlib proves that these points are incident (`cross_orthogonal_left` and its variants). We restate this and add uniqueness.

Lemma 9. *Let $A, B, L, M, P \in \mathbb{P}^2(K)$.*

(a) If $A \neq B$, then A and B lie on `cross A B`. If $L \neq M$, then `cross L M` lies on L and on M.

(b) If $L \neq M$ and P lies on both L and M, then $P =$ `cross L M`. Dually, if $A \neq B$ and both lie on a line L, then $L =$ `cross A B`.

(c) If $A \neq B$, $D \neq E$ and `cross A B` $\neq$ `cross D E`, then the point

`cross (cross A B) (cross D E)`

is collinear with A, B and with D, E, and it is the only such point.

Part (c) connects Mathlib's notion of collinearity with the cross-product construction. It gives a version of Pascal's theorem whose conclusion does not mention `cross` at all.

Theorem 10. *Under the hypotheses of Theorem 1, let P, R, S be points such that P is collinear with A, B and with D, E, the point R with B, C and with E, F, and S with C, D and with F, A. Then $\{P, R, S\}$ is collinear.*

This is the form closest to the informal statement, and to the statements in HOL Light and Mizar (Section 7). The only occurrences of `cross` are the hypotheses that opposite sides are distinct. These hypotheses cannot be dropped. If $AB = DE$, every point of that line is collinear with A, B and with D, E, so P is no longer determined. For example, over $\mathbb{Q}$ take the six affine points $A = (0, 0)$, $B = (1, 0)$, $C = (0, 1)$, $D = (2, 0)$, $E = (3, 0)$, $F = (0, 2)$ on the line pair $xy = 0$. Here $AB = DE$, but $BC \neq EF$ and $CD \neq FA$. Their intersection points are $R = (-3, 4)$ and $S = (0, 1)$. With $P = A$, the points P, R, S are not collinear.

Lemma 11. *For nonzero $v \in K^3$, `OnConic Q (mk K v hv)` holds if and only if $Q(v) = 0$.*

Proof. Every representative has the form λv with $\lambda \neq 0$, and $Q(\lambda v) = \lambda^2 Q(v)$. □

So the conic hypotheses `Q A.rep = 0` of `pascal` do not depend on the representative. `pascal'` restates the theorem with `OnConic`.

Corollary 12 (Pappus)**.** *Let a, b be lines, let A, C, E lie on a and B, D, F lie on b. If consecutive vertices are distinct and opposite sides are distinct lines, then $AB \cap DE$, $BC \cap EF$ and $CD \cap FA$ are collinear.*

Proof. The quadratic form $Q(x) = (a \cdot x)(b \cdot x)$ vanishes on all six points. It is nonzero, since over a field there is a vector z with $a \cdot z \neq 0$ and $b \cdot z \neq 0$. Apply Theorem 1. □

The corollary allows $a = b$. In that case all six points lie on one line, and the hypotheses force every side to be that line, so the hypothesis $AB \neq DE$ fails. The corollary is vacuous there, as expected.

6 Brianchon's theorem

In $\mathbb{P}^2(K)$ with Mathlib's conventions, lines and points are the same objects. A point P lies on a line L when $P \cdot L = 0$, and `cross` computes both the line through two points and the intersection of two lines. Projective duality is therefore literal. Over a field of characteristic different from two, the lines tangent to a nonsingular conic form a nonsingular conic in the dual plane, the *dual conic*. Pascal's theorem, applied in the dual plane, is Brianchon's theorem.

Theorem 13 (Brianchon)**.** *Let Q be a nonzero quadratic form on K^3, and let a, b, c, d, e, f be lines lying on the dual conic $Q = 0$. Suppose consecutive sides are distinct and opposite vertices are distinct points: $a \cap b \neq d \cap e$, $b \cap c \neq e \cap f$, $c \cap d \neq f \cap a$. Then the three diagonals joining opposite vertices are collinear as points of the dual plane. Geometrically this is equivalent to concurrency. Our explicit common-point corollary additionally assumes that the first two diagonals are distinct (the Layer-14 restatement drops this assumption, Section 8.1).*

Proof. The first statement is `pascal'` applied to the lines $a, \ldots, f$. For the second, if three collinear points $L \neq M, N$ of the dual plane are given, then `cross L M` lies on all three lines by Lemma 9. □

This extra assumption is needed only for the chosen cross-product witness.

The formal statement takes the dual conic as its hypothesis. It does not construct the dual conic of a given conic, and it does not define tangency. For a nondegenerate conic over a field of

characteristic $\neq 2$ with Gram matrix M, the dual conic has Gram matrix M^{-1}, and a line is tangent to the conic exactly when it lies on the dual conic. We have not formalised this correspondence, so Theorem 13 is Brianchon's theorem only under this standard identification. Version 2 proves Brianchon's theorem for six distinct tangent lines of a nondegenerate conic in characteristic $\neq 2$ separately, through the polarity of the conic and `pascal'`, in the Layer-15 file (Section 8.1). That proof does not use Theorem 13, and the Gram-matrix description of the dual conic is still not formalised. For arbitrary nonzero dual-plane quadratic forms, including degenerate forms and characteristic two, the formal result is a dual-Pascal generalisation; no tangent interpretation is asserted.

7 Related work

HOL Light. Harrison's `100/pascal.ml` [4] proves Pascal's theorem for a projective conic over $\mathbb{R}$ given by six not-all-zero coefficients. Its statement names the three intersection points x_7, x_8, x_9 by collinearity conditions, as in our Theorem 10. It also assumes that 18 of the 20 triples of vertices are *not* collinear. Affine and circle versions follow; the circle version is the one listed on Wiedijk's page. The proof also uses a bracket condition for six points on a conic, checked with `REAL_RING`. Our approach is therefore close to Harrison's. The differences lie in the generality of the statement. We work over an arbitrary field instead of $\mathbb{R}$. Our hypotheses only require the three intersection points to exist. The 18 non-collinearity conditions are stronger. For example, they require that no three consecutive vertices of the hexagon are collinear. In HOL Light's labelling the hexagon is $x_1x_5x_3x_4x_2x_6$, and the two triples it does not exclude, $\{x_1, x_2, x_3\}$ and $\{x_4, x_5, x_6\}$, are the alternate vertices. So HOL Light's statement does cover the line-pair case of Pappus's theorem, since its conics may be degenerate.

Mizar. Coghetto's article `PASCAL` [2] proves Pascal's theorem in the real projective plane, for a conic given by six real coefficients that are not all zero. The hypotheses form a "Pascal configuration" with 17 non-collinearity conditions, similar to HOL Light's. With the same labelling as HOL Light, it leaves unconstrained the two alternate-vertex triples and one further triple, $\{P_3, P_5, P_6\}$.

Rocq. Magaud and Narboux [6] work in synthetic projective geometry. Their result `hexamy_prop` assumes a strong form of Pappus's theorem and five inequalities between lines. It shows that the "hexamy" property is preserved when the first two vertices are swapped. A hexamy is a hexagon with six pairwise distinct vertices for which the Pascal conclusion holds. It is a statement about axiomatic projective planes, and does not refer to conics defined by equations.

Lean. We know of three public Lean 4 developments, none in Mathlib.

- One proves the theorem for six distinct points on a circle in the Euclidean plane [8], without `sorry`.
- One gives a projective real version whose main theorem depends on a declared `axiom` [5]. Some special cases, such as the standard conic, are proved without it.
- One is an incomplete development for irreducible conics over a field, which relies on a configuration lemma proved with `sorry` [1].

We searched GitHub and the web for others; the search is not exhaustive. Mathlib's list of 100 theorems has no entry for #28 at the version we use [7].

8 Verification

`gate.sh` builds the library and then writes its own temporary file. That file imports the library and issues `#print axioms` for each of the 32 named theorems. The queries are therefore generated by the gate and not read from the repository. The gate flattens Lean's wrapped output lines. It requires exactly one report per named theorem, and accepts only the axioms `propext`, `Classical.choice` and `Quot.sound`. Before building, it rejects source files that contain `sorry`, `admit`, `native_decide`, `axiom`, `#eval`, `run_cmd`, `set_option`, `macro`, `elab`, `syntax`, `notation`, `import Lean`, or related tokens. At commit `726da4d` it prints `PASS (32 theorems, standard axioms only)`. The script needs GNU tools. Its `mktemp -suffix` option is GNU-only, so with the BSD tools of stock macOS it stops with `FAIL: axiom report`.

`tests/tamper.sh` plants six kinds of defect in scratch copies of the source and checks that the gate rejects each one:

- a `sorry`;
- an `axiom` split across two lines;
- an `#eval` that prints a fake axiom report;
- a custom macro;
- an altered incidence hypothesis in `pappus` that is incompatible with the unchanged proof;
- a genuine extra axiom with a long name. The report wraps over several lines, which tests the parser. For this test only, the check for the `axiom` token is switched off in the scratch copy.

All six are rejected.

`code/check_pascal.py` checks the identities with exact rational arithmetic in 300 random trials:

- random points on random conics, which give $\pi = 0$;
- the identity $\pi = -\det V$ on arbitrary points;
- the parametrised standard conic;
- line pairs.

As a control, the script prints π for five perturbed configurations; all five values are nonzero.

The gate is a heuristic, not a sandbox. It trusts the pinned toolchain and Mathlib, and it checks only the named theorems. The step that matters most, comparing the formal statements with Theorem 1, cannot be automated. Section 5 is meant to make that comparison easier. The author has checked each statement in Table 1 against its informal counterpart by hand.

8.1 Layers 14 and 15

Version 2 adds two further statement checks. Both are files of the library in commit `be35ffaa75a324131a710d3287b3d703408ad4a0`, `PascalHexagon/L14.lean` and `PascalHexagon/L15Classical.l` (namespace `PascalHexagon.Classical`). The earlier files are unchanged; the root import file and `gate.sh` gain the new entries. The Layer-15 comparison was first checked as a scratch file and then moved into the library. Below, Layer-14 names are in namespace `PascalHexagon` and Layer-15 names in `PascalHexagon.Classical`.

Layer 14: statements in Mathlib's language and non-vacuity. The main statements of Table 1, except the proof aid `linePair_ne_zero`, are restated with Mathlib-only types and proved from the library theorem through bridges. The restatements unfold `pascalDet`, `quadEval`, `ver`, `coeffs` and `OnConic` into Mathlib's `crossProduct`, dot product, `Matrix.det`, `QuadraticMap.polar` and `Projectivization.rep`. Sixteen theorems restate 21 of the 22 names of Table 1. One theorem covers `pascal`, `pascal'` and `pascal_of_isCollinear`, one covers `brianchon` and `brianchon_concurrent`, and the four names of Lemma 9(a,b) are grouped into two (existence and uniqueness of the line through two points and of the point on two lines). The 22nd name, `linePair_ne_zero`, is not restated. It is a step in the proof of Corollary 12, stated in terms of the library's `linePair`, and it is used in the proof of the restated Pappus theorem. A mechanical check of the elaborated types found no `PascalHexagon` constant. Three restatements are slightly stronger in form:

- `l14_pascal` and `l14_pappus` do not mention `cross` at all. The hypothesis that opposite sides are distinct is written as $\neg(\{A, B, D\}$ and $\{A, B, E\}$ both collinear), which for $A \neq B$, $D \neq E$ is equivalent to `cross A B` $\neq$ `cross D E` (bridge `l14b_cross_ne_iff`).
- `l14_brianchon` proves that the three diagonals have a common point *without* the hypothesis that the first two diagonals are distinct (bridge `l14b_common_point`). It still takes the dual conic as a hypothesis, as in Section 6.

Five witness theorems prove every hypothesis on explicit configurations over $\mathbb{Q}$ and that the conclusion is non-degenerate:

statement	witness and non-degeneracy proved
Pascal (Thm. 1, 10)	six points on $x^2+y^2=z^2$, 15 vertex pairs distinct; Pascal points pairwise distinct
Pappus (Cor. 12)	A, C, E on $y=0$, B, D, F on $x=0$; Pappus points pairwise distinct
Brianchon (Thm. 13)	six tangents of the unit circle; diagonals pairwise distinct; common point $(7:-9:1)$
Prop. 2–Rem. 7	$\pi=-51$ at an explicit point; $c\neq 0$, $Q\neq 0$, $M\neq 0$ symmetric; vertices and Pascal points distinct
Lemmas 8, 9, 11	a collinear and a non-collinear triple; distinct points and lines; a point on and a point off the conic

`l14_apply_eq_coeffs` is an identity with no hypotheses and needs no witness. No Mathlib-only statement turned out false or needed an extra hypothesis. Tangency of the six Brianchon lines in the witness was checked numerically, not in Lean. The example in Section 5 showing that $AB \neq DE$ cannot be dropped is not formalised. With the Layer-14 file added, the gate checks 59 theorems (32 + 27) and passes, and the tampering suite rejects all six defects. With the Layer-15 file added as well, the gate checks 147 theorems (59 + 88) and passes, and the tampering suite again rejects all six defects. In that commit the GNU-only `mktemp` call in `gate.sh` is replaced, so the gate also passes with the BSD tools of stock macOS; `tests/tamper.sh` still needs GNU `sed`.

Layer 15: comparison with the classical statements. Another AI agent collected the classical statements from the literature, with citations (Pascal's *Essay pour les coniques*, Brianchon's 1806 memoir and the modern projective form in [3], Ch. 9). These were Pascal's theorem, its five-point (tangent) form, Pappus's theorem, the converse (Braikenridge–Maclaurin), Brianchon's theorem for a nondegenerate conic in characteristic $\neq 2$, its polarity duality with Pascal, and its pentagon form. A writer AI model was given only these statements; it saw neither the code nor this paper, and its transcript was audited (a transcript audit in the verification repository). It wrote them in Lean with Mathlib's `Projectivization`, with lines as 2-dimensional subspaces of K^3, and with its own definitions of a nondegenerate conic ("not equivalent to a form in fewer variables", with at least one point), of the tangent at A (a line through A that meets the conic in no other point) and of the polar line. This gave 20 statements, 12 theorems and 8 well-formedness obligations, all type-checked with `sorry`. A comparer then proved each one, copied verbatim, from the library, with no added hypothesis and no change to the library, and classified it. All 20 are proved; `#print axioms` reports only the three standard axioms for each:

- *Covered* (6): Pascal's theorem (`cov_Pascal1_a`, from `pascal'`) and Pappus's theorem (`cov_Pascal2_b`, from `pappus`; the library file inlines the one-line proof of `pappus`, `pascal'` applied to `linePair`, so that the tampering test that replaces `pappus` is still rejected by the axiom check), with their three obligations and the obligation that XY is a line. In both cases the library statement is strictly more general. It allows any nonzero form (resp. $a = b$ and vertices at $a \cap b$) and requires only distinct consecutive vertices and distinct opposite sides.

- *Narrower* (1): Brianchon's theorem for six distinct tangent lines. Missing from the library: the link between tangent lines and the dual conic, which Theorem 13 takes as a hypothesis. NEW PROOF `cov_Brianchon1_a` (characteristic $\neq 2$) closes the gap through the polarity of the conic, applied to `pascal'`. It does not use `brianchon`.

- *Out of scope of the paper, now proved* (13), all NEW PROOFS:

  - a nondegenerate conic contains no three collinear points, in any characteristic (`cov_Setting_noThreeCollinear`);
  - the five-point form with the tangent at A, in any characteristic (`cov_Pascal2_a`, with its obligation). The proof applies Lemma 3 over $K[X]$ and specialises $X = 0$;
  - the converse: six distinct points, no three collinear, whose Pascal points are collinear, lie on a nondegenerate conic, in any characteristic (`cov_Pascal3_a`, with its obligation). The proof uses Proposition 2;
  - the polarity (`cov_Brianchon2_a_`X for $X =$ `polarIsLine`, `tangent`, `vertex`, `incidence`, `diagonal`). The tangent equals the polar in any characteristic; the others are stated and proved in characteristic $\neq 2$;
  - the obligation of Brianchon's theorem (opposite vertices are distinct), and the pentagon form with its obligation (`cov_Brianchon2_b`).

- *Writer statement false as written*: none.

The Layer-15 file reproduces the writer's definitions and statements verbatim. Relative to the scratch file, the library version changes the namespace, header and imports, removes two linter options, and inlines the proof of `pappus` in `cov_Pascal2_b`, as described above.

9 Limitations and further work

- *Tangents.* The projective statement requires consecutive vertices to be distinct. The limiting cases, in which a side is replaced by the tangent to the conic, are not part of the main development, since Mathlib has no definition of the tangent line to a conic. With the Layer-15 writer's definition (a line through A meeting the conic only at A), the five-point form and the pentagon form of Brianchon's theorem are proved in the Layer-15 file (Section 8.1), for nondegenerate conics only. Other limiting cases are not formalised.
- *The affine and Euclidean statements.* Wiedijk's list uses an affine or circle statement. Deriving one from Theorem 10 needs a bridge between Mathlib's affine notion `Collinear` in $\mathbb{R}^2$ and `IsCollinear` in $\mathbb{P}^2(\mathbb{R})$. We have not formalised it.
- *Converse.* The converse of Pascal's theorem, that a hexagon whose opposite sides meet in collinear points lies on a conic, follows from the same identity. It is not part of the main development. The Layer-15 file proves that six distinct points, no three collinear, whose Pascal points are collinear lie on a nondegenerate conic in the writer's sense (`cov_Pascal3_a` in namespace `PascalHexagon.Classical`, Section 8.1).
- *Tangency and the dual conic.* Formalising the correspondence between tangent lines and the dual conic would make Theorem 13 a statement about tangent lines directly. The tangent-line form itself is proved in the Layer-15 file through the polarity (characteristic $\neq 2$), but the correspondence with the dual conic is not.
- *Mathlib.* Contributing the result to Mathlib would require a general notion of conic and faster proofs of Proposition 2.

Acknowledgements and use of AI

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